

■ 1.4.4 Making Replacements

Making an *assignment* like $x = 3$ tells *Mathematica* to replace x by 3 *whenever* x appears. Often, you will need to make replacements in a more controlled fashion. For example, you may just need to replace x by 3 in a *particular* expression.

You can do this in *Mathematica* by typing $expr /. x \rightarrow 3$. This takes the expression $expr$ and replaces all the occurrences of x in $expr$ by 3. You must type $\rightarrow$ as a pair of characters with no space in between. You can think of $x \rightarrow 3$ as representing a “replacement” in which “ x goes to 3”. The operator $/.$ is also typed as a pair of characters with no space in between. You can think of it as representing evaluation of the expression “at the point” specified by the replacement.

This replaces x by $1+a$ in the expression $1+x^2$.

```
In[1]:= 1 + x^2 /. x -> 1+a
```

```
Out[1]= 1 + (1 + a)^2
```

Making a replacement for x using $/.$ has no effect on the value of x .

```
In[2]:= x
```

```
Out[2]= x
```

You can do several replacements at once by putting them in a list.

```
In[3]:= (x + y) (x - y)^2 /. {x -> 3, y -> 1 - a}
```

```
Out[3]= (3 - (1 - a))^2 (4 - a)
```

$x = value$	define a value for x which will always be used
$x = .$	remove any value defined for x
$expr /. x \rightarrow value$	replace x by $value$ in the expression $expr$
$expr /. \{x \rightarrow xval, y \rightarrow yval\}$	perform several replacements

Assignments and replacements.

You can mix assignments and replacements.

This assigns a value to the symbol t .

```
In[4]:= t = 1 + x^2
```

```
Out[4]= 1 + x^2
```

This finds the value of t , and then replaces x by 2 in it.

```
In[5]:= t /. x -> 2
```

```
Out[5]= 5
```

This finds the value of t for a different value of x .

```
In[6]:= t /. x -> 5a
```

```
Out[6]= 1 + 25 a2
```

This finds the value of t when x is replaced by Pi , and then evaluates the result numerically.

```
In[7]:= t /. x -> Pi //N
```

```
Out[7]= 10.8696
```